Problem

The ordinary difference amplifier for fixed-gain operation is shown in Fig. 5.81(a). It is simple and reliable unless gain is made variable. One way of providing gain adjustment without losing simplicity and accuracy is to use the circuit in Fig. 5.81(b). Another way is to use the circuit in Fig. 5.81(c). Show that:

(a) for the circuit in Fig. 5.81(a),

$$\frac{v_o}{v_i} = \frac{R_2}{R_1}$$

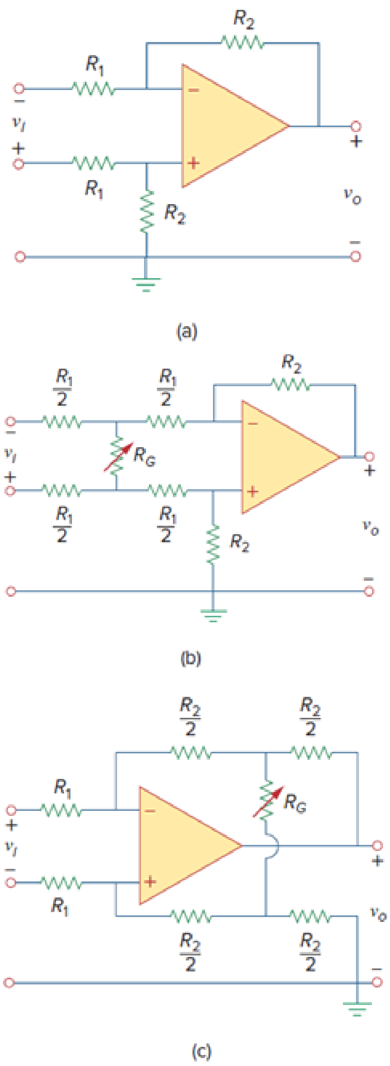
(b) for the circuit in Fig. 5.81(b),

$$\frac{v_o}{v_i} = \frac{R_2}{R_1}\frac{1}{1+\frac{R_1}{2R_G}}$$

(c) for the circuit in Fig. 5.81(c),

$$\frac{v_o}{v_i} = \frac{R_2}{R_1}\left(1+\frac{R_2}{2R_G}\right)$$

Figure. 5.81:



Step-by-step solution

Step 1 of 11

(a)

Refer to Figure 5.81(a) in the textbook.

Draw the circuit diagram with node voltage representation.

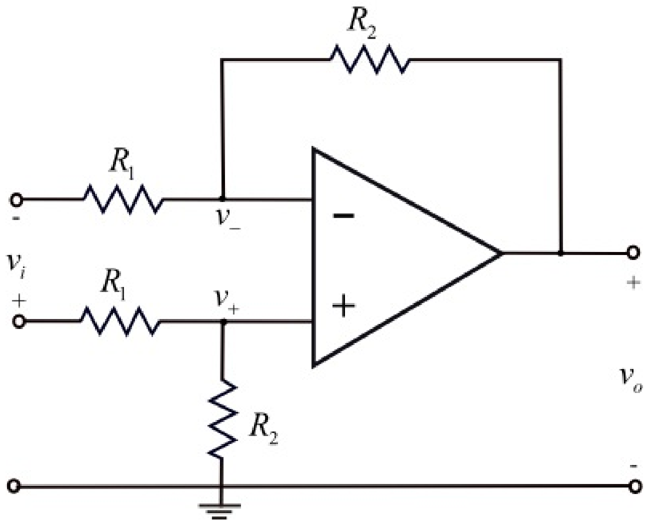


Figure 1

Step 2 of 11

Assume that the op-amp is ideal.

Therefore,

$$v_+ = v_-$$

Write the Kirchhoff's current law equation at non inverting node.

$$\frac{v_+ - v_i}{R_1} + \frac{v_+}{R_2} = 0$$

$$v_+\left(\frac{1}{R_1} + \frac{1}{R_2}\right) = \frac{v_i}{R_1}$$

$$v_+ = v_i\left(\frac{R_2}{R_1 + R_2}\right) \quad \ldots\ldots (1)$$

Write the Kirchhoff's current law equation at inverting node.

$$\frac{v_- - v_o}{R_2} + \frac{v_-}{R_1} = 0$$

$$\left(\frac{1}{R_2} + \frac{1}{R_1}\right)v_- = \frac{v_o}{R_2}$$

$$\left(\frac{R_1 + R_2}{R_2 R_1}\right)v_+ = \frac{v_o}{R_2} \qquad (\text{Since, } v_- = v_+)$$

Substitute $v_i\left(\frac{R_2}{R_1 + R_2}\right)$ for v_+.

$$\left(\frac{R_1 + R_2}{R_2 R_1}\right)\left(\frac{R_2}{R_1 + R_2}\right)v_i = \frac{v_o}{R_2}$$

$$\left(\frac{1}{R_1}\right)v_i = \frac{v_o}{R_2}$$

$$\frac{v_o}{v_i} = \frac{R_2}{R_1}$$

Therefore, the output gain, $\frac{v_o}{v_i}$ is $\boxed{\frac{R_2}{R_1}}$.

Step 3 of 11

(b)

Refer to Figure 5.81 (b) in the text book.

Draw the circuit diagram with node voltage representation.

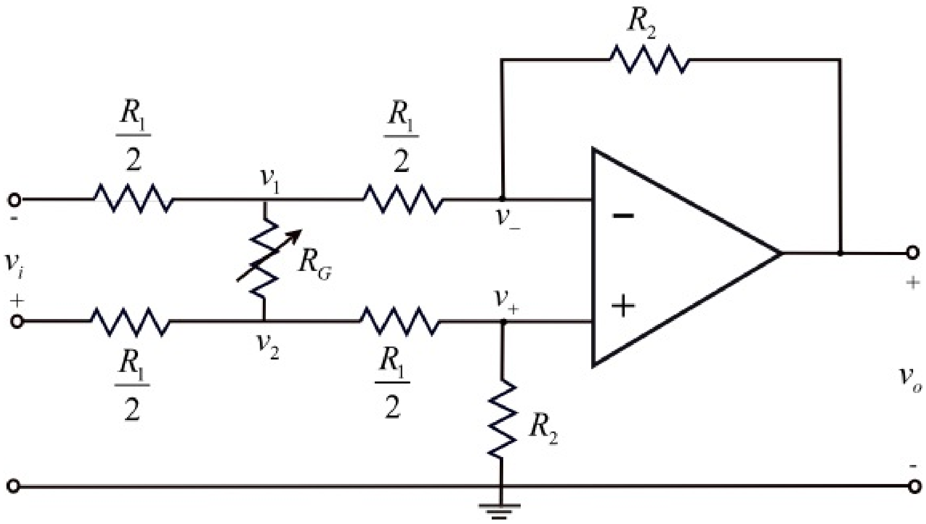


Figure 2

Step 4 of 11

Write the Kirchhoff's current law equation at node, v_1.

$$\frac{v_1}{\frac{R_1}{2}} + \frac{v_1 - v_2}{R_G} + \frac{v_1 - v_-}{\frac{R_1}{2}} = 0$$

$$\frac{v_1 + v_1 - v_+}{\frac{R_1}{2}} + \frac{v_1 - v_2}{R_G} = 0 \qquad (\text{Since, } v_+ = v_-)$$

$$2v_1 - v_+ + \frac{R_1}{2R_G}(v_1 - v_2) = 0 \text{ ...... (2)}$$

Write the Kirchhoff's current law equation at node, v_2.

$$\frac{v_2 - v_i}{\frac{R_1}{2}} + \frac{v_2 - v_+}{\frac{R_1}{2}} + \frac{v_2 - v_1}{R_G} = 0$$

$$\frac{2v_2 - v_i - v_+}{\frac{R_1}{2}} + \frac{v_2 - v_1}{R_G} = 0$$

$$2v_2 - v_i - v_+ + \frac{R_1}{2R_G}(v_2 - v_1) = 0 \text{ ...... (3)}$$

Step 5 of 11

Subtract equation (2) from equation (3).

$$2v_2 - v_i - v_+ + \frac{R_1}{2R_G}(v_2 - v_1) - 2v_1 + v_+ - \frac{R_1}{2R_G}(v_1 - v_2) = 0$$

$$v_i = \frac{2R_1}{2R_G}(v_2 - v_1) + 2(v_2 - v_1) = 0$$

$$\frac{v_i}{2} = \left(1 + \frac{R_1}{2R_G}\right)(v_2 - v_1)$$

$$v_2 - v_1 = \frac{v_i}{2}\left(\frac{1}{1 + \frac{R_1}{2R_G}}\right) \quad \ldots\ldots (4)$$

For the difference amplifier,

$$v_o = \frac{R_2}{\frac{R_1}{2}}(v_2 - v_1)$$

$$v_2 - v_1 = \frac{R_1}{2R_2}v_o \quad \ldots\ldots (5)$$

Step 6 of 11

Substitute equation (5) in equation (4).

$$\frac{R_1}{2R_2}v_o = \frac{v_i}{2}\left(\frac{1}{1 + \frac{R_1}{2R_G}}\right)$$

$$\frac{v_o}{v_i} = \frac{R_2}{R_1}\left(\frac{1}{1 + \frac{R_1}{2R_G}}\right)$$

Therefore, the output gain, $\frac{v_o}{v_i}$ is $\boxed{\frac{R_2}{R_1}\left(\frac{1}{1 + \frac{R_1}{2R_G}}\right)}$.

Step 7 of 11

(c)

Refer to Figure 5.81 (c) in the text book.

Draw the circuit diagram with node voltage representation.

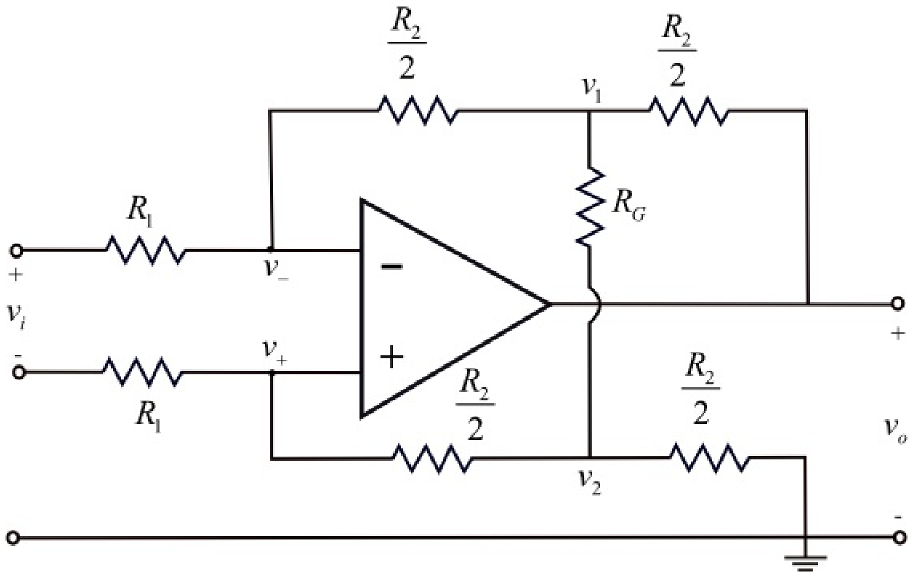


Figure 3

Step 8 of 11

Write the Kirchhoff's current law equation at node, v_-.

$$\frac{v_- - v_i}{R_1} + \frac{v_- - v_1}{\frac{R_2}{2}} = 0$$

$$\frac{v_i - v_-}{R_1} = \frac{v_- - v_1}{\frac{R_2}{2}}$$

$$v_i - v_- = \frac{2R_1}{R_2}v_- - \frac{2R_1}{R_2}v_1 \quad \text{...... (6)}$$

Write the Kirchhoff's current law equation at node, v_+.

$$\frac{v_+}{R_1} + \frac{v_+ - v_2}{\frac{R_2}{2}} = 0$$

$$v_+ = \frac{2R_1}{R_2}v_2 - \frac{2R_1}{R_2}v_+ \quad \text{...... (7)}$$

Step 9 of 11

Add equation (7) with equation (6).

$$v_i - v_- + v_+ = \frac{2R_1}{R_2}v_- - \frac{2R_1}{R_2}v_1 + \frac{2R_1}{R_2}v_2 - \frac{2R_1}{R_2}v_+$$

Since, $v_+ = v_-$.

$$v_i - v_- + v_- = \frac{2R_1}{R_2}v_- - \frac{2R_1}{R_2}v_1 + \frac{2R_1}{R_2}v_2 - \frac{2R_1}{R_2}v_-$$

$$v_i = -\frac{2R_1}{R_2}v_1 + \frac{2R_1}{R_2}v_2$$

$$v_i = \frac{2R_1}{R_2}(v_2 - v_1)$$

$$v_2 - v_1 = \frac{R_2}{2R_1}v_i \text{ ...... (8)}$$

Step 10 of 11

Write the Kirchhoff's current law equation at node, v_1.

$$\frac{v_1 - v_o}{\frac{R_2}{2}} + \frac{v_1 - v_2}{R_G} + \frac{v_1 - v_-}{\frac{R_2}{2}} = 0$$

$$v_1 - v_o + \frac{R_2}{2R_G}(v_1 - v_2) + v_1 - v_- = 0 \text{ ...... (9)}$$

Write the Kirchhoff's current law equation at node, v_2.

$$\frac{v_2}{\frac{R_2}{2}} + \frac{v_2 - v_+}{\frac{R_2}{2}} + \frac{v_2 - v_1}{R_G} = 0$$

$$v_2 + v_2 - v_+ + \frac{R_2}{2R_G}(v_2 - v_1) = 0 \text{ ...... (10)}$$

Step 11 of 11

Subtract equation (9) from equation (10)

$$v_2 + v_2 - v_+ + \frac{R_2}{2R_G}(v_2 - v_1) - \left(v_1 - v_o + \frac{R_2}{2R_G}(v_1 - v_2) + v_1 - v_- \right) = 0$$

$$2v_2 - 2v_1 + \frac{R_2}{2R_G}(v_2 - v_1) + v_o - \frac{R_2}{2R_G}(v_1 - v_2) = 0$$

$$2v_2 - 2v_1 + \frac{2R_2}{2R_G}(v_2 - v_1) + v_o = 0$$

$$2(v_2 - v_1)\left(1 + \frac{R_2}{2R_G}\right) = -v_o \quad \text{...... (11)}$$

Substitute $\frac{R_2}{2R_1} v_i$ for $v_2 - v_1$ in equation (11).

$$2\left(\frac{R_2}{2R_1} v_i\right)\left(1 + \frac{R_2}{2R_G}\right) = -v_o$$

$$\left(\frac{R_2}{R_1} v_i\right)\left(1 + \frac{R_2}{2R_G}\right) = -v_o$$

$$\frac{v_o}{v_i} = -\left(\frac{R_2}{R_1}\right)\left(1 + \frac{R_2}{2R_G}\right)$$

Therefore, the output gain, $\frac{v_o}{v_i}$ is $\boxed{-\left(\frac{R_2}{R_1}\right)\left(1 + \frac{R_2}{2R_G}\right)}$.